

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12383854
Kind Code	B2
Date of Patent	August 12, 2025
Inventor(s)	Louis; Dominique et al.

Clamp system and filter arrangement and methods

Abstract

A filter system and clamp system for a gas turbine air inlet uses a single clamp system that allows for securing in place both a first element and a second element on the grid of the gas turbine air inlet. The clamp system can be released to allow for services of the first filter element, while the second filter element remains in sealing engagement with the grid.

Inventors:	Louis; Dominique (Brussels, BE), Van Gelder; Wim V. (Kessel-Lo, BE), Nicols; Benjamin M. (Bierbeck, BE), Yadav; Rakesh K (Lakeville, MN), Ronneau; Oliver (Chapelle-Lez-Herlaimont, BE)
Applicant:	Donaldson Company, Inc. (Bloomington, MN)
Family ID:	78830600
Assignee:	Donaldson Company, Inc. (Bloomington, MN)
Appl. No.:	18/000788
Filed (or PCT Filed):	June 04, 2021
PCT No.:	PCT/US2021/036006
PCT Pub. No.:	WO2021/248070
PCT Pub. Date:	December 09, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20230211271 A1	Jul. 06, 2023

Related U.S. Application Data

Publication Classification

Int. Cl.: **B01D46/42** (20060101); **B01D46/00** (20220101); **B01D46/02** (20060101); **B01D46/121** (20220101); **F02C7/055** (20060101)

U.S. Cl.:

CPC **B01D46/4227** (20130101); **B01D46/0005** (20130101); **B01D46/023** (20130101); **B01D46/121** (20220101); B01D2265/027 (20130101); B01D2271/02 (20130101); B01D2279/60 (20130101); F02C7/055 (20130101)

Field of Classification Search

CPC: B01D (46/12); B01D (46/121); B01D (46/4227); B01D (46/0005); B01D (46/023); B01D (2265/027); B01D (2271/02); B01D (2271/022); B01D (2271/025); B01D (2271/027); B01D (2279/60); F02C (7/055)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7334490	12/2007	Morse et al.	N/A	N/A
7931723	12/2010	Cuvelier	N/A	N/A
8105409	12/2011	Mann et al.	N/A	N/A
8425644	12/2012	Sundvik et al.	N/A	N/A
8673040	12/2013	Handley et al.	N/A	N/A
8951321	12/2014	Gillingham et al.	N/A	N/A
9623356	12/2016	Gorman	N/A	N/A
10486094	12/2018	Katare et al.	N/A	N/A
2010/0192528	12/2009	Mann	55/482	B01D 46/58
2010/0229514	12/2009	Ayshford	55/482	B01D 46/0005
2010/0251678	12/2009	Mann	55/483	B01D 46/121
2014/0096493	12/2013	Kelmartin et al.	N/A	N/A
2018/0221801	12/2017	Eyers	N/A	F02C 7/052
2023/0175717	12/2022	Yadav	55/385.1	A01K 1/0082

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2012/164087	12/2011	WO	N/A
2016/045730	12/2015	WO	N/A

OTHER PUBLICATIONS

International Search Report and Written Opinion for PCT/US2021/036006 (Sep. 17, 2021). cited by applicant

Background/Summary

(1) This application is a National Stage Application of PCT International Patent application No. PCT/US2021/036006, filed Jun. 4, 2021, which claims the benefit of priority to U.S. Provisional patent application Ser. No. 63/035,412, filed Jun. 5, 2020, and U.S. Provisional patent application Ser. No. 63/140,020, filed Jan. 21, 2021, and which applications are incorporated herein by reference. To the extent appropriate, a claim of priority is made to each of the above disclosed applications.

TECHNICAL FIELD

(1) This disclosure relates to air filter systems. In certain applications, it concerns air filters for use in air intake of gas turbine systems and a clamp arrangement. Methods are also provided.

BACKGROUND

(2) Although the present disclosure may be applied in a variety of applications, it was developed for use with gas turbine filter systems. Gas turbine systems are useful in generating electricity. These types of systems are convenient in that they can be constructed quickly and they produce fewer harmful emissions than coal or oil. Gas turbines utilize air for combustion purposes. The combustion air needs to be clean. To ensure clean air for combustion, air filters have been used to clean the air taken into the gas turbine system.

(3) Filters are used to purify the air intake for gas turbines. An example gas turbine air intake system is disclosed in U.S. Pat. No. 8,673,040, incorporated herein by reference.

(4) The air filters used in these intake systems need to be periodically removed and replaced. It is desirable to find a convenient system for allowing for the quick and easy servicing of these air filters. In particular, it is desirable to have a system in which a single clamp system can be used for both the first filter and second filter, and allow the second filter to remain sealed in place while the first filter is being serviced.

SUMMARY

(5) This disclosure provides a solution to the problem in the prior art. For example, the solution provided herein allows for a single clamp that seals both a first filter and second filter to the support grid of an intake air system of a gas turbine and also allows for keeping the second filter in sealing engagement with the support grid while releasing the first filter for replacement. The clamp can also selectively release the second element for periodic removal and replacement.

(6) In a first aspect, a clamp system for releasably attaching a filter assembly to a support grid is provided; the clamp system includes: (a) an elongated support member with first and second opposite open ends, a first slide channel adjacent the first end; and a second slide channel adjacent the second end; (b) a first clamp slidably mounted in the first open end; (c) a second clamp slidably mounted in the second open end; (d) a first slide pin arrangement slidably mounted in the first slide channel between a locked and a released position; (e) a second slide pin arrangement slidably mounted in the second slide channel between a locked and a released position; (f) an elongated omega clamp removably mounted to cover the support member; the omega clamp having first and second spaced through slots; (i) the first through slot slidably accommodating a portion of the first slide pin arrangement; (ii) the second through slot slidably accommodating a portion of the second slide pin arrangement; (g) first and second pivot rods extending through and connecting the omega clamp and the support member; the first and second pivot rods each having a hinge point in the support member and a pivot axis along a length of the support member between the first and

second open ends of the support member; wherein, when the first and second slide pin arrangements are slid to the released position, the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support member.

(7) The clamp system can further include a separate maintenance tool constructed and arranged to engage each of the first and second pivot rods and apply an axial force against the omega clamp.

(8) The maintenance tool can include (a) a head defining a recess sized to engage the first and second pivot rods, and a cam surface on a portion of the head opposite of the recess; and (b) a grasping handle extending from the head.

(9) Each of the first and second pivot rods includes a barrel nut on a portion of the pivot rods opposite of the hinge point.

(10) Each of the first clamp and second clamp includes: (a) a slide rod sized to be slidably received within one of the first and second open ends of the support member; (b) a post secured to and perpendicular to the slide rod; and (c) a flange secured to a distal portion of the post and parallel to and facing the slide rod to form a gasket holding recess therebetween.

(11) Each of the first and second slide pin arrangements includes: (a) a pin slide rod sized to slide within one of the first and second slide channels; (b) a support secured to and perpendicular to the pin slide rod; (c) a pin flange secured to a distal portion of the support and parallel to the pin slide rod with an interior side facing the pin slide rod and an opposite exterior side; and (d) pin projecting from the exterior side and sized to slide within one of the first and second slots in the omega clamp.

(12) The omega clamp comprises: (a) a base with a pair of legs extending perpendicular to the base; (i) the base defining the first and second through slots; (ii) the base having an exterior side and an opposite interior side; (b) each of the legs being bent at a non-zero angle remote from the base and away from each other; the legs and the interior side of the base defining an open channel; and (c) first and second pin slider keepers secured to the interior side of the base and overlapping a portion of the first and second through slots.

(13) The omega clamp comprises: (a) a base with a pair of legs extending perpendicular to the base; (i) the base defining the first and second through slots; (ii) the base having an exterior side and an opposite interior side; (b) each of the legs being bent at a non-zero angle remote from the base and away from each other; the legs and the interior side of the base defining an open channel; and (c) first and second pin slider keepers secured to the interior side of the base and overlapping a portion of the first and second through slots; (i) the first and second pin slider keepers being sized to slidably receive one of the pin flanges to put the first and second slide pin arrangements in the locked positions; and wherein when the first and second slide pin arrangements are slid to the released position, each of the pin flanges are slid to be free of the first and second pin slider keepers, and the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support member.

(14) In another aspect, a clamp system for releasably attaching a filter assembly to a support grid is provided comprising: (a) a support member; (b) first and second clamps slidably mounted at opposite ends of the support member; (c) a first and second slide pin arrangements slidably mounted a portion of the support member; (d) an omega clamp removably mounted to cover the support member and slidably accommodating a portion of the first and second slide pin arrangement; (e) first and second pivot rods extending through and connecting the omega clamp and the support member; the first and second pivot rods each having a hinge point in the support member and a pivot axis along a length of the support member between opposite ends of the support member; and wherein, when the first and second slide pin arrangements are slid to a released position, the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support member.

(15) In another aspect, a filter system is provided comprising: (a) a support grid having a frame with opposite first and second sides; (b) a first filter element oriented along the first side of the

support grid; (c) a second filter oriented on the support grid directly across and opposing the first filter; and (d) a single clamp system releasably securing both the first filter element and the second filter element to the frame.

(16) The clamp system comprises: (a) a support member; (b) first and second clamps slidably mounted at opposite ends of the support member; (c) a first and second slide pin arrangements slidably mounted a portion of the support member; (d) an omega clamp removably mounted to cover the support member and slidably accommodating a portion of the first and second slide pin arrangement; (e) first and second pivot rods extending through and connecting the omega clamp and the support member; the first and second pivot rods each having a hinge point in the support member and a pivot axis along a length of the support member between opposite ends of the support member; and wherein, when the first and second slide pin arrangements are slid to a released position, the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support member.

(17) The clamp system includes: (a) an elongated support member with first and second opposite open ends, a first slide channel adjacent the first end; and a second slide channel adjacent the second end; (b) a first clamp slidably mounted in the first open end; (c) a second clamp slidably mounted in the second open end; (d) a first slide pin arrangement slidably mounted in the first slide channel between a locked and a released position; (e) a second slide pin arrangement slidably mounted in the second slide channel between a locked and a released position; (f) an elongated omega clamp removably mounted to cover the support member; the omega clamp having first and second spaced through slots; (i) the first through slot slidably accommodating a portion of the first slide pin arrangement; (ii) the second through slot slidably accommodating a portion of the second slide pin arrangement; (g) first and second pivot rods extending through and connecting the omega clamp and the support member; the first and second pivot rods each having a hinge point in the support member and a pivot axis along a length of the support member between the first and second open ends of the support member; wherein, when the first and second slide pin arrangements are slid to the released position, the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support member.

(18) The filter system can include (a) the first filter includes a perimeter frame with a gasket thereon; (b) the second filter includes a perimeter frame with a gasket thereon; and (c) a seal is formed with the first filter perimeter gasket being between and against the first and second clamps and the perimeter frame of the second filter.

(19) A seal is formed with the second filter perimeter gasket being between and against the omega clamp and the support grid frame.

(20) In one example, the first filter comprises a pocket filter; and the second filter comprises a pocket filter.

(21) In another aspect, a method of servicing a filter arrangement in a gas turbine air inlet system is provided; the system having a support grid having a frame with opposite first and second sides; a first filter element releasably sealed to the first side of the support grid; and a second filter releasably sealed to the first side of the support grid directly across and opposing the first filter; the method comprising: (a) releasing a single clamp holding both of the first filter element and second filter element in place in sealing engagement on the support grid; (b) while the second filter element remains in sealing engagement on the support grid, removing the first filter element; and (c) providing a replacement first filter element and engaging the single clamp to seal the replacement first filter element to the support grid.

(22) The method can further comprise: (a) before the step of providing a replacement first filter element, releasing another part of the single clamp to remove the second filter element; and (b) providing a replacement second filter element and engaging the single clamp to seal the replacement second filter element to the support grid.

(23) The method can further include using the single clamp as variously characterized above.

(24) In another aspect, a clamp system for releasably attaching a filter assembly to a support grid is provided. The clamp system comprising a plurality of clamp rods; a first elongated support member with openings, each opening being sized to fit over one of the clamp rods; a first slide plate slidably mounted in the first support member; the first slide plate having openings alignable with the openings in the first support member; the first slide plate having an end handle; a second elongated support member with openings, each opening in the second support member sized to fit over one of the clamp rods; and a plurality of handgrips, each handgrip constructed and arranged to engage one of the clamp rods and apply an axial force against the second support member.

(25) The clamp system can further include a second slide plate slidably mounted in the second support member; the second slide plate having openings alignable with the openings in the second support member; and the second slide plate having an end handle.

(26) In many examples, the handgrips each comprises: a head defining a recess sized to engage one of the clamp rods, and a cam surface on a portion of the head opposite of the recess; and a grasping handle extending from the head.

(27) The first support member may have a rectangular cross-sectional wall with opposite open ends, and the openings in the first support member extend through opposing sides of the wall; and the second support member may have a rectangular cross-sectional wall with opposite open ends, and the openings in the second support member extend through opposing sides of the wall of the second support member.

(28) In one or more examples, the first slide plate openings have a narrowed region and a wider region.

(29) In example embodiments, the second slide plate openings have a narrowed region and a wider region.

(30) There can be two clamp rods and two handgrips.

(31) In another aspect, a filter system is provided comprising: a support grid having a frame with opposite upstream and downstream sides; a first filter element oriented along the upstream side of the support grid; a second filter element oriented on the support grid on the downstream side directly across and opposing the first filter element; and a single clamp system releasably securing both the first filter element and the second filter element to the frame.

(32) The second filter element may have a second element gasket pressed against the upstream side of the support grid; and the first filter element may have a first element gasket pressed against the clamp system.

(33) In example embodiments, the clamp system comprises: one or more clamp rods extending from the support grid; a first elongated support member holding a first slide plate and positioned over the clamp rods; the first support member being positioned against the second element gasket; a second elongated support member positioned over the clamp rods and against the first element gasket, such that the first element gasket is positioned between and against the first support member and second support member; and one or more handgrips removably secured to the clamp rods and applying an axial force against the second support member.

(34) Many embodiments further comprise a second slide plate being held by the second support member and positioned over the clamp rods.

(35) In examples, there are a plurality of first filter elements on the upstream side of the support grid; and the first filter elements comprise pocket filter elements or V-bank elements.

(36) In examples, there are a plurality of second filter elements on the downstream side of the support grid, and the second filter elements comprise V-bank filter elements.

(37) In another aspect, a method of servicing a filter arrangement in a gas turbine air inlet system is provided; the system having a support grid having a frame with opposite upstream and downstream sides; a first filter element releasably sealed to the upstream side of the support grid; and a second filter element releasably sealed to the upstream side of the support grid directly across and opposing the first filter element; the method comprising: releasing a single clamp assembly holding

both of the first filter element and second filter element in place in sealing engagement on the support grid; while the second filter element remains in sealing engagement on the support grid, removing the first filter element; and providing a replacement first filter element and engaging the single clamp assembly to seal the replacement first filter element to the support grid.

(38) The method may further comprise, before the step of providing a replacement first filter element, releasing another part of the single clamp assembly to remove the second filter element; and providing a replacement second filter element and engaging the single clamp assembly to seal the replacement second filter element to the support grid.

(39) In one or more example methods, the step of engaging the single clamp assembly to seal the replacement first filter element to the support grid includes grasping a handgrip and rotating the handgrip from an open position to a closed position to apply an axial force against the clamp assembly.

(40) The step of engaging the single clamp assembly can include, after rotating the handgrip to apply an axial force against the clamp assembly, moving a first slide plate to lock the second filter element into a sealed position against the support grid.

(41) The method can further include, after the step of moving a first slide plate to lock the second filter element into a sealed position against the support grid, moving a second slide plate to lock the first filter element into a sealed position; and rotating the handgrip back to the open position and removing the handgrip.

(42) The method can include the single clamp assembly as variously characterized above.

(43) A variety of examples of desirable features or methods are set forth in the description that follows.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a perspective, exploded view of a filter system for use with a gas turbine air system, the FIG. depicting a first filter element, a second filter element, a portion of the gas turbine support grid, and a clamp system;

(2) FIG. 2 is a perspective view of the system of FIG. 1, with the second filter element secured in place, and the first filter element being removed from the support grid;

(3) FIG. 3 is a perspective view of a portion of the system of FIG. 1 with both the first filter element and second element in place against the support grid and the clamp system engaged securing both the first filter element and second filter element;

(4) FIG. 4 is an exploded, perspective view of the clamp system used in FIGS. 1-3;

(5) FIG. 5 is an enlarged front view of a portion of the clamp system during one step of servicing the second filter elements;

(6) FIG. 6 is a perspective view showing one step of servicing the second filter elements;

(7) FIG. 7 is an enlarged view showing a cross-section of both the first filter element and second filter element in sealing engagement against the support grid of the gas turbine air inlet system; and

(8) FIG. 8 is a schematic, perspective view showing the gasket of the first filter element, with the remaining part of the first filter element hidden to enhance clarity, and how it is held in place by the clamp system.

(9) FIG. 9 is a perspective view of another embodiment of a gas turbine air inlet system, constructed in accordance with principles of this disclosure:

(10) FIG. 10 is the system of FIG. 9, but smaller, to show reference to additional drawings;

(11) FIG. 11 is an enlarged view of the portion shown at A of FIG. 10;

(12) FIG. 12 is an enlarged view of the portion shown at B of FIG. 10,

(13) FIG. 13 is a top plan view of the system of FIG. 9;

- (14) FIG. 14 is a cross-sectional view of a portion shown at cross section C-C of FIG. 13;
- (15) FIG. 15 is a perspective view showing a clamp system used in the air inlet system of FIG. 9;
- (16) FIG. 16 is another perspective view of the clamp system of FIG. 15;
- (17) FIG. 17 is a top plan view of the clamp system of FIG. 15;
- (18) FIG. 18 is an exploded perspective view of the clamp system of FIG. 15;
- (19) FIG. 19 is another exploded perspective view of the clamp system of FIG. 15;
- (20) FIG. 20 is a perspective view of a clamp rod used in another embodiment of a clamp system shown in FIGS. 24-39;
- (21) FIG. 21 is a perspective view of another clamp rod used in clamp system of FIGS. 24-39;
- (22) FIG. 22 is a perspective view of a portion of the air inlet system using the clamp system of FIGS. 24-39;
- (23) FIG. 23 is a view similar to FIG. 22, but showing another part of the clamp system of FIGS. 24-39;
- (24) FIG. 24 is a perspective view of the clamp assembly used for the air inlet system of FIGS. 22 and 23;
- (25) FIG. 25 is an enlarged view of section A of FIG. 24;
- (26) FIG. 26 is a front view of the clamp system of FIG. 24;
- (27) FIG. 27 is a cross-sectional view of section 27-27 of FIG. 26;
- (28) FIG. 28 is a perspective view of the air inlet system during one step of applying the clamp system of FIGS. 24-27;
- (29) FIG. 29 is a perspective view showing another step of applying the clamp system of FIGS. 24-27;
- (30) FIG. 30 is a cross-sectional view of a portion of the clamp system shown in FIG. 29;
- (31) FIG. 31 is a perspective view showing another step of applying the clamp system of FIGS. 24-27;
- (32) FIG. 32 is a perspective view of a portion of the air inlet system having the clamp system of FIGS. 24-27;
- (33) FIG. 33 is an enlarged perspective view of the portion shown at Y of FIG. 32;
- (34) FIG. 34 is an enlarged perspective view of the portion shown at Z of FIG. 32;
- (35) FIG. 35 is a perspective view showing the clamp system of FIGS. 24-27;
- (36) FIG. 36 is another perspective view of clamp system of FIG. 35;
- (37) FIG. 37 is a top plan view of the clamp system of FIG. 35;
- (38) FIG. 38 is an exploded perspective view of the clamp system of FIG. 35;
- (39) FIG. 39 is another exploded perspective view of the clamp system of FIG. 35;
- (40) FIG. 40 is an exploded perspective view of a variation of the system of FIGS. 9-19;
- (41) FIG. 41 is a perspective view of a portion of the system of FIG. 40;
- (42) FIG. 42 is an enlarged cross-sectional view of a portion of FIG. 41;
- (43) FIG. 43 is a top plan view of the system of FIG. 40;
- (44) FIG. 44 is an enlarged view of a portion of FIG. 43;
- (45) FIG. 45 is an exploded perspective view of a variation of the system of FIGS. 9-19;
- (46) FIG. 46 is a perspective view of a portion of the system of FIG. 45;
- (47) FIG. 47 is an enlarged cross-sectional view of a portion of FIG. 46;
- (48) FIG. 48 is a top plan view of the system of FIG. 45;
- (49) FIG. 49 is an enlarged view of a portion of FIG. 48;
- (50) FIG. 50 is an exploded perspective view of a variation of the system of FIGS. 9-19;
- (51) FIG. 51 is a perspective view of a portion of the system of FIG. 50;
- (52) FIG. 52 is an enlarged cross-sectional view of a portion of FIG. 51;
- (53) FIG. 53 is an enlarged cross-sectional view of a portion of FIG. 51;
- (54) FIG. 54 is an enlarged cross-sectional view of a portion of FIG. 51;
- (55) FIG. 55 is a perspective view of the system of FIG. 51;

(56) FIG. 56 is a top plan view of the system of FIG. 55;
(57) FIG. 57 is an enlarged view of a portion of FIG. 56; and
(58) FIG. 58 is a schematic view of an example system utilizing one or more of the filtration systems and components of FIG. 1-57.

DETAILED DESCRIPTION

(59) FIG. 1 shows a portion of a gas turbine air inlet system having a first filter element 20 and a second filter element 22. FIG. 1 also shows a portion of a first filter element 20 both vertically above and below the first element 20 shown in its entirety. Similarly, the second filter element 22 also shows a portion of another second element 22 both vertically above and vertically below it.
(60) A variety of different types of filter elements can be used. In this particular example, the first filter element 20 is a pocket filter element, such as shown and described in U.S. Pat. No. 7,931,723, incorporated herein by reference. The second filter element 22 can also be a pocket filter, although a number of different variations are possible. Each of the elements 20, 22 has filter media 21, 23, respectively. The first filter element 20 could also be a v-bank filter.
(61) In FIG. 1, a portion of a support grid 24 having a frame 26 is shown. The support grid 24 will include a matrix of several, for example hundreds, of rectangular grids with openings 28. Each opening 28 accommodates one of the second filters 22 and one of the first filters 20. In this example, the support grid 26 has opposite first and second sides 29, 30. The first side 29 is generally an upstream side, while the side 30 is a downstream side. When installed, the first filter element 20 and the second filter element 22 will be oriented on the support grid 24 in opposition to each other, such that the first filter element 20 is on the upstream side 24, while the second filter element 22 is on the downstream side 30.

The Example Embodiment of the Clamp System of FIGS. 1-8

(62) Also visible in FIG. 1 is a clamp system 32. The clamp system 32 is shown at an upper portion of FIG. 1 as being exploded from the support grid 24, while the clamp system 32 at a lower portion of the FIG. is shown secured to the support grid 24. The clamp system 32 is a single clamp system that releasably secures the both the first filter element 20 and the second filter element 22 to the frame 26 of the support grid 24.

(63) The first filter element 20 includes a perimeter frame 34 all along each side of the first element 20. The perimeter frame includes a first side 34a (FIG. 7) facing away from the filter media 21 and an opposite second side 34b facing toward the filter media 21. Secured to the perimeter frame 34 on the first side 34a is a gasket 36 (FIGS. 7 and 8). Gasket 36 is secured to the frame 34 on the first side 34a facing away from the filter media 21 and is along the entire perimeter of the first element 20. The gasket 36 can be made from a variety of materials, including a soft foam.

(64) The second element 22 includes a perimeter frame 38 as can be seen in FIG. 1 the perimeter frame 38 is along the entire perimeter of the second element 22. The perimeter frame 38 includes a first side 38a facing away from the filter media 23, and an opposite second side 38b (FIG. 7) facing toward the filter media 23. Secured to the perimeter frame 38 of the second element 22 is a gasket 40. The gasket 40 is not on the side 38a that is visible in FIG. 1 but can be seen in the cross-section in FIG. 7 at 38b. The gasket 40 is on a portion of the frame 38 that faces a remaining part of the second element 22. As can be appreciated from reviewing FIG. 1 and FIG. 7, the gasket 40 for the second filter element 22 is compressed against the first or upstream side 29 of the frame 26 of the support grid 24. The gasket 36 of the first filter element 20 is then pressed against the frame 38 (first side 38a of the frame 38) of the second element 22. The clamp 32 holds the first filter element 20 and second filter element 22 in place in sealing engagement against the support grid 24.

(65) In general, in operation of the gas turbine air inlet system, air to be filtered enters the first filter element 20, in which particulate and debris are removed by passing through the filter media 21. The filtered air then flows through the second element 22 by flowing from inside of the second element 22 through the media 23 to the outside on the downstream side, and then proceeds to the gas turbine. In this arrangement, the first filter element 20 is located generally on an upstream side

of the support grid **24**, while the second element **22** is primarily located on the downstream side of the support grid **24**, although the seal made with the gasket **40** of the second element **22** is along the first or upstream side **29** of the support grid **24**.

(66) In reference now to FIG. **4**, an example clamp system **32** is illustrated in an exploded view.

(67) The clamp system **32** is used for releasably attaching the filter assembly, comprising the first element **20** and the second element **22** to the support grid **24**. The clamp system **32** allows for using the single system **32** to hold both the first filter element **20** and the second filter element **22** in place in sealing engagement on the support grid **24**. While the second filter element **22** remains in sealing engagement on the support grid **24**, the clamp system **32** can be released to allow for removal and replacement of the first filter element **20**. The clamp system **32** also allows for periodic removal and replacement of second element **22**. Typically, the first filter elements **20** need removal and replacement at more frequent intervals than the second elements **22**.

(68) The clamp system **32** includes an elongated support member or tube **45**. The support tube **45** has first and second opposite open ends **46**, **47**. The support tube **45** also includes a first slide channel **48** adjacent to the first end **46**, and a second slide channel **49** adjacent the second end **47**. The first and second slide channels **48**, **49** are formed by openings in the tube **45** spaced from the ends **46**, **47**. As can be seen, there is a first enclosure **50** and a second enclosure **51**, such that the region between the ends of the enclosure **50**, **51** and the open ends **46**, **47** is an enclosed open through channel.

(69) The tube **45** further includes a through hole **52** through the enclosure **50** and a remaining portion of the tube **45** and a through hole **53** through the enclosure **51** and a remaining portion of the tube **45**. The through holes **52**, **53** receive projections or pegs **55**, **56** (FIG. **1**) extending from the frame **26** of the support grid **24**. This allows for mounting of the tube **45** on and against the frame **26** of the grid **24**.

(70) The clamp system **32** further includes first and second pivot rods **58**, **59**. The first and second pivot rods **58**, **59** each has a swivel type hinge point **60**, **61** held within the support tube **45** within a respective one of the enclosures **50**, **51**. The first and second pivot rods **58**, **59** pivot along a pivot axis **62** which is along a length of the support tube **45** between the first and second open ends **46**, **47**. Arrows **63**, **64** show in general the direction of pivoting of the rods **58**, **59** about their hinge points **60**, **61**. The purpose for this pivoting is described further below.

(71) In addition, in FIG. **4**, each of the first and second pivot rods **58**, **59** includes a barrel nut **66**, **67** on a portion of the pivot rod **58**, **59** that is remote from or opposite of the hinge points **60**, **61**. The barrel nuts **66**, **67** allow for engagement with a maintenance tool **70**, described further below.

(72) The clamp system **32** further includes a first clamp **74** slidably mounted in the first open end **46** and a second clamp **75** slidably mounted in the second open end **47**. Each of the first clamp **74** and second clamp **75** includes a slide rod **76**, **77** sized to slide within one of the first and second open ends **46**, **47** of the support tube **45**. An extension or post **78**, **79** is secured to and perpendicular to the respective slide rod **76**, **77**. A flange **80**, **81** is secured to a distal portion of the respective post **78**, **79** and is parallel to and facing the slide rod **76**, **77** to form a gasket holding recess **82**, **83** therebetween. The first clamp and second clamp **74**, **75** releasably hold the first filter elements **20** in place in sealing engagement.

(73) The clamp system **32** further includes a first slide pin arrangement and a second slide pin arrangement **86**, **87**. The first slide pin arrangement **86** is slidably mounted in the first slide channel **48** between a locked and a released position. The second slide pin arrangement **87** is slidably mounted in the second slide channel **49** between a locked and released position. In the locked position, the second filter element **22** is secured in place against the grid **24**. In the unlocked position, the second filter element **22** can be removed from the grid **24** for servicing (replacement). This is described further below.

(74) Each of the first and second slide pin arrangements **86**, **87** includes a pin slide rod **89**, **90** sized to slide within a respective one of the first and second slide channels **48**, **49**. The slide channels **48**,

49 are within the enclosures **50, 51**. Each of the first and second slide pin arrangements **86, 87** further includes a respective support **91, 92** secured to and perpendicular to the pin slide rod **89, 90**. (75) A pin flange **93, 94** is secured to a distal portion of the respective support **91, 92** and parallel to the pin slide rod **89, 90** with an interior side of the pin flange **93, 94** facing the respective pin slide rod **89, 90**. The pin flange **93, 94** also has an opposite exterior side. Each of the first and second slide pin arrangements **86, 87** further includes a pin **95, 96**. The pins **95, 96** project from the exterior side of the pin flange **93, 94** and is sized to slide within first and second slots **98, 99** of an omega clamp **100**.

(76) The clamp system **32** includes elongated omega clamp **100**. The omega clamp **100** is removably mounted to cover the support tube **45**. The omega clamp **100** includes a base **102** with a pair of legs **103, 104** extending perpendicular to the base **102**. The base **102** defines the first and second through slots **98, 99**, which accommodate the pins **95, 96**. As can be seen in FIG. 4, the slots **98, 99** are spaced from the opposite ends of the omega clamp **100** and also spaced from each other.

(77) The base **102** had an exterior side **106** (FIG. 7) and an opposite interior side **107** (FIG. 7). Each of the legs **103, 104** is bent at a non-zero angle remote from the base **102** and away from each other to form pressing flanges **108, 109**. The non-zero angles can include a range of, for example, 30-90° of the pressing flanges **108, 109** from the portion of the legs **103, 104** that are perpendicular to the base **102**, but many embodiments are possible. The legs **103, 104** and the interior side **107** of the base **102** define an open channel **110**.

(78) The omega clamp **100** further includes first and second pin slider keepers **112, 113**. In FIG. 4, the pin slider keepers **112, 113** are shown in broken lines, as they are on the interior side **107** of the base **102**. The slider keepers **112, 113** overlap a portion of the first and second through slots **98, 99**.

(79) The first and second pin slider keepers **112, 113** are sized to slidably receive a respective one of the pin flanges **93, 94** to put the first and second slide pin arrangements **86, 87** in the locked positions. When the first and second slide pin arrangements **86, 87** are slid to the released position, the omega clamp **100** is free to be pivoted with the first and second pivot rods **58, 59** along the pivot axis **62** and away from covering the support tube **45**. That is, when the first and second pin slide arrangements **86, 87** are slid to the released position, each of the pin flanges **93, 94** is slid to be free of the first and second pin slider keepers **112, 113**, and the omega clamp **100** is free to be pivoted along the pivot axis **62** in the direction of arrow **63, 64**.

(80) When assembled, the first and second pivot rods **58, 59** extend through and connect the omega clamp **100** and the support tube **45**. The base **102** of the omega clamp **100** includes through holes **114, 115**, which accommodate the pivot rods **58, 59**.

(81) Still in reference to FIG. 4, the clamp system **32** includes a separate maintenance tool **70**. The maintenance tool **70** is constructed and arranged to engage each of the first and second pivot rods **58, 59** and apply an axial force against the omega clamp **100** during servicing of the second filter elements **22**.

(82) The maintenance tool **70** includes a head **118** defining a recess **119** sized to engage the first and second pivot rods **58, 59**. The head **118** includes a cam surface **120** on a portion of the head **118** opposite of the recess **119**. The maintenance tool **70** also includes a grasping handle **121** sized to be gripped by a human hand. The grasping handle **121** extends from the head **118**.

(83) As can be seen in FIG. 5, the maintenance tool **70** can be used so that the recess **119** is placed on the barrel nut **66, 67** of a respective one of the pivot rods **58, 59**. The maintenance tool **70** is then rotated or pivoted about the barrel nut **66, 67**, and this moves the camming surface **120** against the omega clamp **100**. Specifically, the camming surface **120** can be moved to provide a force against the base **102** of the omega clamp **100**. This then applies pressure to the frame **38** of the second element **22** to press against the gasket **40** of the second element **22**.

(84) FIG. 3 shows a portion of the system when the first element **20** and second element **22** are in place, and the clamp system **32** is engaged securing both elements **20, 22** in place. The pin **95** of

the first slide pin arrangement **86** can be seen in a locked position, in which it is in the position to the most outside portion of the first slide **98**. It is being held by the first slider keeper **112**. The first pivot rod **58** can be seen extending through the hole **114** in the omega clamp **100**. The flange **80** of the first clamp **74** can be seen pressing against the frame **34** of the first filter element **20**.

(85) An example method of servicing the filter arrangement is now explained. In reference to FIG. 2, the first step is to slide the first and second clamps **74**, **75** in an outward direction from the rest of the assembly, such that they are slid away from each other and toward the outside. This can be seen in FIG. 2, in which the clamps **74**, **75** have been slid away so that the flanges **80**, **81** are away from what would be the frame **34** of the first filter element **20**. This step allows the second filter element **22** to remain in sealing engagement on the support grid **24**, but releases the first element **20** so that it can be removed. A new, replacement first element **20** can be provided. The new first element **20** is placed so that the perimeter frame **34** of the first element **20** holding the gasket **36** can be placed against the first side of the perimeter frame **38a**. The replacement filter element **20** can be axially pressed against the frame **38a** of the second element **22**, and then the first and second clamps **74**, **75** can again be slid inwardly toward each other so that the flanges **80**, **81** are resting against the perimeter frame **34** on the second side **34b** of the perimeter frame **34** that is opposite from the first side **34a** on which the gasket **36** is located. The flanges **80**, **81** help to hold the first element **20** in place, with a seal formed against the frame **38** (second side **38b**) of the second element **22**. See FIG. 7.

(86) From time to time, it will also be desirable to service and replace the second elements **22**. During this maintenance step, the gas turbine inlet system is shut down. First, the first filter element **20** is removed, by the steps above. That is, the first and second clamps **74**, **75** are moved away from each other to the outside, which allows for release and removal of the first element **20**. Next, the maintenance tool **70** is used to press the omega clamp **100** against the perimeter frame **38** of the second element **22** and compress the gasket **40** of the second element **22** to a maximum displacement. The maintenance tool **70** is used by placing the recess **119** against the barrel nut **66**, **67** and pivoting the maintenance tool **70** so that the cam surface **120** presses against the base **102** of the omega clamp **100**.

(87) Next, the pins **95**, **96** are moved or slid toward each other to the inside within their respective slots **98**, **99**. This releases the pin flanges **93**, **94** from the pin slider keepers **112**, **113**. This also releases the omega clamp **100** from the support tube **45**.

(88) Reference is now made to FIG. 6. The omega clamp **100** can then be pulled and lifted out of the way of the second filter element **22**. In particular, it can be pivoted in the direction of arrows **63**, **64** (FIG. 4) and along the pivot axis **62** either upwardly as shown in FIG. 6 or downwardly, along with pivot rods **58**, **59**. This then allows access to the second element **22**, so that the second element **22** can be removed from the support grid **24**.

(89) A new second element **22** is provided and placed in the opening **28**. The gasket **40** is pressed against the first side **29** of the support grid **24**. The omega clamp **100** is then pivoted over the perimeter frame **38** of the second element **22** so that the gasket **40** of the second element **22** is pressed between and against the perimeter frame **38** (i.e., second side **38b**) of the second element **22** and the support grid **24**. The first element **20** is then placed directly across and in opposition to the second element **22** so that the gasket **36** of the first element **20** is pressed against the frame **38** (i.e., first side **38a**) of the second element **22**. The first and second clamps **74**, **75** are then moved to the inside so that the flanges **80**, **81** press against the second side **34b** and hold the perimeter frame **34** of the first element **20**.

The Example Embodiment of FIGS. 9-19

(90) In FIG. 9, a gas turbine air inlet system is shown at **200**. The gas turbine air inlet system **200** is similar to that as described in U.S. Pat. No. 8,673,040, incorporated herein by reference. There are rows and columns of filter elements arranged on a tube sheet or support grid, as described previously with respect to support grid **24**. The support grid **24** includes frame **26** having upstream

side **29** and opposite downstream side **30**. Each opening in the support grid **24** accommodates at least one first filter element **202** on the upstream side **29** and one second filter element **204** on the downstream side **30**. As can be seen in FIG. **9**, the support grid **24** includes a matrix of several, for example hundreds, of first filter elements **202** on the upstream side **29** and second filter elements **204** on the downstream side **30**.

(91) As depicted, the first filter element **202** and second filter element **204** are typically oriented on the support grid **24** directly across from each other and covering one of the openings **28** in the support grid **24**.

(92) Many different types of filter elements can be used. In the example shown, the first filter elements **202** comprise pocket filter elements **205**, while the second filter elements **204** comprise v-bank filter elements **206**. Example pocket filter elements are described in U.S. Pat. No. 7,931,723, incorporated herein by reference. Example v-bank filter elements are described in U.S. Pat. No. 8,425,644, U.S. Ser. No. 10/486,094, U.S. Pat. Nos. 7,334,490, 9,623,356, and 8,951,321, each of which is incorporated herein by reference. In other embodiments, the first filter elements **202** could also be v-bank filter elements.

(93) The first filter element **202** and the second filter element **204** each has a perimeter gasket **201**, **203**, which is constructed and arranged to be axially pressed to form a seal.

(94) As with the embodiment of FIGS. **1-8**, a single clamp system **210** is used to releasably secure both a first filter element **202** and second filter element **204** to the frame **26** of the support grid **24**.

(95) In reference now to FIGS. **16-19**, clamp system **210** is depicted. The clamp system **210** releasably attaches the filter assembly, comprising the first filter element **202** and the second filter element **204** to the support grid **24**. The clamp system **210** includes a plurality of clamp rods **212**, **214**.

(96) In the example shown, each clamp system **210** includes two clamp rods **212**, **214**. The clamp rods **212**, **214** will generally be non-removably secured to the frame **26** to project therefrom. Each of the rods **212**, **214** has a cylindrical base **216** extending from a grid engaging portion to about halfway of the length of the rod **212**, **214**. The other half of the rod **212**, **214** extending to a free (terminal) end has a narrowed section **218** being a handgrip-engaging section **218** shaped to receive one of the handgrips **220**, **222**. The narrowed section **218** comprises a projection **219** extending from a side or lateral portion thereof.

(97) The clamp system **210** further includes a first elongated support member or tube **224**. The first support tube **224** can be many different shapes having a variety of cross-sections. In the particular, non-limiting example shown, the first support tube **224** includes a generally rectangular cross-sectional wall **226** with opposite open ends **227**, **228**. The wall **226** is positioned on a flange plate **230** supporting it along the length.

(98) The support tube **224** can have a single (FIGS. **45-49**) or a plurality of openings. In the non-limiting example shown in FIGS. **9-19**, there are two openings **232**, **234**. In this case, the openings **232**, **234** match up and are in axial alignment with the clamp rods **212**, **214**. Each opening **232**, **234** is sized to receive or be fit over one of the clamp rods **212**, **214**. Openings **232**, **234** also extend through the flange plate **230**.

(99) The clamp system **210** further includes a first slide plate **236**. The first slide plate **236** is removably and slidably mounted in the first support tube **224**. In the example shown, the first slide plate **236** is insertable through the open end **228** and is within the rectangular wall **226**.

(100) The first slide plate **236** has openings shown at **238**, **240**. The openings **238**, **240** are alignable with the openings **232**, **234** of the first support tube **224**. The openings **238**, **240** can be shaped to assist with locking the second filter element **204** in sealing engagement with the grid **24**, as further describe below. In the example shown, the openings **238**, **240** have a narrowed region **242** and a wider region **244**.

(101) The first slide plate **236** further includes an end handle **246**. The end handle **246** will project out of the first support tube **224** extending from the open end **228**, when the first slide plate **236** is

positioned within the first support tube **224**. The end handle **246** can be used to move the slide plate **236** between the locked and unlocked position. The end handle **246** is generally perpendicular to a remaining portion of the first slide plate **236**.

(102) The clamp system, **210** further includes a second elongated support profile, member, or tube **250**. The second support tube **250** further includes a plurality of openings shown at opening **252** and **254**. Each opening **252**, **254** is arranged and sized to fit over one of the clamp rods **212**, **214**.

(103) The second support tube **250** has a rectangular cross-sectional wall **256**. The wall **256** has opposite open ends **257**, **258**. The openings **252**, **254** extend through opposing sides of the wall **256**.

(104) Also shown in FIG. **18** is a mounting plate **260** with posts **262**, **264** projecting therefrom. Mounting bases **266**, **268** receive the posts **262**, **264**. The mounting bases **266**, **268** are secured to the rods **212**, **214**. The assembly including the plate **260** with posts **262**, **264** and mounting bases **266**, **268** are secured to or can be integral with the support grid **24**. Many variations are possible.

(105) The hand grips **220**, **222** are analogous to the maintenance tool **70** described previously. The hand grips **220**, **222** are constructed and arranged to engage one of the clamp rods **212**, **214** and apply an axial force against the support tube **250**.

(106) The hand grips **220**, **222** each include a head **270**. The head **270** defines a recess **272** sized to engage the grip engaging section or narrower section **218** of the rods **212**, **214**. Each head **270** further includes a cam surface **275** in the form of a flattened area and on a portion of the head **270** that is opposite of the recess **272**. A grasping handle **276** is sized for comfortable gripping by a human hand extends from the head **270**.

(107) In FIG. **19**, it can be seen how each hand grip **220**, **222** has its head **270** split into head pieces **277**, **278**. Between the head pieces **277**, **278** is a receiving slot **280**. The receiving slot **280** is sized to fit around the hand grip engaging section **218** of the rods **212**, **214**. Each head piece **277**, **278** includes recessed **272**, which will hook onto the projection **219** of the rods **212**, **214**.

(108) In use, the hand grips **220**, **222** can be fit around the rods **212**, **214** such that the recess **272** in the head **270** will engage the projection **219**, and the hand grip **220** can be pivoted from an unlocked position shown at **282** in FIGS. **9** and **17** to a locked **284** also shown in FIG. **9**. The unlocked position includes the cam surface **272** being out of engagement with the second support tube **250** and the handles **276** extending away from the support grid **24**. The locked position **284** include the cam surface **274** being against the second support tube **250** and the handles **276** generally pointing to each other and almost generally parallel to the support grid **24**.

(109) In the system of FIGS. **9-19**, the second filter element **204** is initially placed through one of the openings in the support grid **24** such that it is on the downstream side **30**, with its gasket **203** on the upstream side **29** and against the frame **26**. The first support tube **224** along with the first slide plate **236**, is then placed over the rods **212**, **214**, such that the flange plate **230** of the support tube **224** is against a header frame of the second filter element **204**. The slide plate **236** can then be moved or slid by pulling on end handle **246** in a direction laterally outwardly from the second filter element **204**. The gasket **203** is not yet compressed.

(110) Next, the second support profile or tube **250** is placed over the rods **212**, **214**, and the hand grips **220**, **222** are loosely fit onto the clamp rods **212**, **214**. The first filter element **202** is then installed by placing the header frame of the first filter element **202** underneath the second support tube **250**. The hand grips **220**, **222** are then rotated from a position where the handles **276** are generally perpendicular to the frame **26**, to a position in which the handles **276** are generally parallel to the frame **26**. This will compress both of the gaskets **201**, **203** of the first filter element **202** and second filter element **204**. Next, the slide plate **236** is moved or slid by grasping the end handle **246** and moving it in a direction toward the elements **202**, **204**, which keeps the second filter element **204** clamped with the gasket **203** in the compressed state and allows for online changing of the first filter elements **202**.

(111) To service the filter arrangement of FIGS. **9-19**, the process is done in reverse, by releasing

the clamp system **210** holding both the first filter element **202** and second filter element **204** in place and in sealing engagement on the support grid **24**. While the second filter element **204** remains in sealing engagement, the first filter element **202** is removed. A replacement first filter element **202** is provided and the single clamp system **210** is engaged to seal the replacement first filter element **202** to the support grid **24**.

(112) The step of engaging the single clamp system **210** can include grasping the hand grips **220**, **222** and rotating the hand grips **220**, **222** from an open position to a closed position to apply an axial force against the clamp system **210**. After rotating the hand grips **220**, **222**, the first slide plate **236** can be moved to lock the second filter element **204** into a sealed position against the support grid **24**.

(113) Periodically, the second filter element **204** can also be replaced. This is done by, after removing the first filter element **202**, sliding the slide plate **236** to an unlocked or released position by moving it laterally (sideways) away from the elements **202**, **204**. This releases the seal of the gasket **203** on the second element **204**. The first support tube **224** can then be removed, which allows the second filter element **204** to be removed from the frame **26**. A replacement second element **204** is then provided and inserted into the opening in the frame **26**, with the gasket **203** against the upstream side **29** of the frame **26**. The first support tube **224** is again mounted onto the rods **212**, **214**. The second support tube **250** is then mounted onto the rods **212**, **214**, and the first filter element **202** is oriented so that the gasket **201** is between the first support tube **224** and second support tube **250**. The hand grips **220**, **222** are rotated such that the cam surface **274** is moved against the second support tube **250** to apply axial force to the first and second support tubes **224**, **250** compressing the gaskets **201**, **203**. The first slide plate **236** is then moved into a locked position, sliding in a direction toward the elements **202**, **204**.

The Example Embodiment of FIGS. **20-39**

(114) In reference now to FIGS. **20-39**, an alternative embodiment for the clamp system **210** is provided, depicted generally at **201'**. The difference between the clamp system **210** and clamp system **210'** is the existence of a second slide plate **300** (FIG. **38**). The second slide plate **300** is held by the second support tube **250** and is positioned over the clamp rods **212**, **214**. The second slide plate **300** includes an end handle **302** projecting generally perpendicular to a remaining portion thereof. In this system, after the hand grips **220**, **222** are rotated into the locked position, both the first slide plate **236** and second slide plate **300** are moved or slid into the locked position, in which the first slide plate **236** secures that gasket **203** of the second filter element **204** into a sealed position, and the second slide plate **300** secures the gasket **201** of the first filter element **202** into a sealed position, after that, the hand grips **220**, **222** are removable from the overall system, and both the first filter element **202** and the second filter element **204** are secured in place. The motion to move the first slide plate **236** and second slide plate **300** to the locked position includes moving the slide plates **236**, **300** laterally outwardly, in a direction away from the filter elements **202**, **204**. In the example shown, the end handles **246**, **302** are on opposite sides of the elements **202**, **204** and are moved away from each other when being moved to the locked position. In other embodiments, this motion could be reversed—that is, instead of moving the slide plates **236**, **300** laterally away from each other to the locked position, they could be designed oppositely so that moving the slide plates **236**, **300** laterally toward each other results in the locked position.

(115) When the system of FIGS. **20-39** is serviced, the second slide plate **300** is slid by moving it inwardly toward a direction of the first filter element **202** and second filter element **204**. This will allow the second support tube **250** to be lifted or pivoted and remove the first filter element **202**. A replacement first filter element **202** then can then be provided and oriented with the gasket **201** between the first support tube **224** and the second support tube **250**. The hand grips **220**, **222** can then again be engaged with the rods **212**, **214** and rotated to compress the second support tube **250** against the gasket **201** of the first filter element **202**. The second slide plate **300** can then be moved to lock the first filter element **202** into a sealed position.

(116) Occasionally, the second filter element **204** will also be replaced, and, after removing the first filter element **202** as described above, the second filter element **204** can be removed and replaced, as described in detail with respect to the embodiment of FIGS. **9-19**.

The Example Embodiment of FIGS. **40-44**

(117) FIGS. **40-44** show a variation of the embodiment of FIGS. **9-19**, with the same reference numerals being used to show similar parts. In this embodiment, instead of having the second support member **250**, there is a second support member **250'** having the same profile shape as the first elongated support member **224**.

The Example Embodiment of FIGS. **45-49**

(118) FIGS. **45-49** show a variation of the embodiment of FIGS. **9-19**, with the same reference numerals being used to show similar parts. In this embodiment, instead of having a pair of handgrips **220**, **222**, there is a single handgrip **220'** centered on the support members **224**, **250**. There is also only a single clamp rod **212'** fixed to the grid **26**, horizontally centered for each filter **204**, **205**.

The Example Embodiment of FIGS. **50-57**

(119) FIGS. **50-57** show a variation of the embodiment of FIGS. **9-19**, with the same reference numerals being used to show similar parts.

(120) In this embodiment, both the first filter element **202** and the second filter element **204** are V-bank filter elements **206**. There is greater horizontal and vertical spacing between each element **206**, than in the embodiment of FIGS. **9-19**, one of the purposes being to accommodate additional handgrips, as explained below.

(121) The system **200** includes handgrips **223** along the lateral sides of each element **206** to help hold the elements **206** in place on the frame **26**. Extending from vertically members of frame **26** are connector rods **213** that engage with handgrips **223**. There are also clamp members **285**, **286** to help clamp the elements **206** in place against the element frame headers **287**, **288**.

The Example Embodiment of FIG. **58**

(122) The air filtration systems above can be used in a variety of environments and systems. For example, the air filtration systems described above can be used to filter incoming air to prevent airborne viruses. Such applications can include livestock environments, such as swine farms, poultry, and other livestock in which airborne viruses could transmit.

(123) Static positive pressure air filtration systems using small turbine units can be used. Filter elements with filtration ratings from MERV 14 to HEPA can be applied to capture relevant airborne viruses. One unit can providing 250,000 CFM with 90 filter elements, and multiple units can be configured to meet overall air flow requirements based on livestock population to be housed.

(124) In the prior art, a negative pressure air filtration with under 0.2 inch of water of pressure differential for sizing of the system is used. In these prior art systems, to provide high efficiency filtration and required flow, a significantly large systems are required. The systems described in this disclosure are about $\frac{1}{3}$ of the size of prior art solutions with significantly reduced maintenance and operational resources.

(125) Using the air filtration systems of FIGS. **1-57**, a positive pressure filtration system is used, which results in a smaller system, and has advantages of ease of maintenance and better security during changeover. It will have up to 1.0 inch of water positive pressure differential, and less than 300 filter units.

(126) In reference now to FIG. **58**, a system is shown at **300** having a livestock barn **302**. The barn **302** can be for a variety of livestock such as swine **304**, or can be for poultry (e.g., chickens, turkeys), sheep, goats, cattle/bovines, horses, etc.

(127) The system **300** includes filtration system **306**, which can include any of the systems described above in connections with FIGS. **1-57**. The filtration system **306** draws in air from the outside atmosphere at an inlet region **308**. The air is filtered to remove contaminants, including viruses, and the filtered air exits through an outlet region **310** in the interior **312** of the barn **302**. As

shown here, the outlet region **310** is adjacent a ceiling **314** of the livestock area in the interior **312** of the barn **302**. Interior air from the interior **312** of the barn **302** is exhausted to the outer atmosphere from the barn **302** through an exhaust shutter **316**.

(128) The above provides example principles of operation. Many embodiments can be made using these principles.

Claims

1. A clamp system for releasably attaching a filter assembly to a support grid; the clamp system comprising: (a) an elongated support tube with first and second opposite open ends, a first slide channel adjacent the first end; and a second slide channel adjacent the second end; (b) a first clamp slidably mounted in the first open end; (c) a second clamp slidably mounted in the second open end; (d) a first slide pin arrangement slidably mounted in the first slide channel between a locked and a released position; (e) a second slide pin arrangement slidably mounted in the second slide channel between a locked and a released position; (f) an elongated omega clamp removably mounted to cover the support tube; the omega clamp having first and second spaced through slots; (i) the first through slot slidably accommodating a portion of the first slide pin arrangement; (ii) the second through slot slidably accommodating a portion of the second slide pin arrangement; (g) first and second pivot rods extending through and connecting the omega clamp and the support tube; the first and second pivot rods each having a hinge point in the support tube and a pivot axis along a length of the support tube between the first and second open ends of the support tube; wherein, when the first and second slide pin arrangements are slid to the released position, the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support tube.
2. The clamp system of claim 1 further including a separate maintenance tool constructed and arranged to engage each of the first and second pivot rods and apply an axial force against the omega clamp.
3. The clamp system of claim 2 wherein the maintenance tool comprises: (a) a head defining a recess sized to engage the first and second pivot rods, and a cam surface on a portion of the head opposite of the recess; and (b) a grasping handle extending from the head.
4. The clamp system of claim 1 wherein each of the first and second pivot rods includes a barrel nut on a portion of the pivot rods opposite of the hinge point.
5. The clamp system of claim 1 wherein each of the first clamp and second clamp includes: (a) a slide rod sized to be slidably received within one of the first and second open ends of the support tube; (b) a post secured to and perpendicular to the slide rod; and (c) a flange secured to a distal portion of the post and parallel to and facing the slide rod to form a gasket holding recess therebetween.
6. The clamp system of claim 1 wherein each of the first and second slide pin arrangements includes: (a) a pin slide rod sized to slide within one of the first and second slide channels; (b) a support secured to and perpendicular to the pin slide rod; (c) a pin flange secured to a distal portion of the support and parallel to the pin slide rod with an interior side facing the pin slide rod and an opposite exterior side; and (d) pin projecting from the exterior side and sized to slide within one of the first and second slots in the omega clamp.
7. The clamp system claim 1 wherein the omega clamp comprises: (a) a base with a pair of legs extending perpendicular to the base; (i) the base defining the first and second through slots; (ii) the base having an exterior side and an opposite interior side; (b) each of the legs being bent at a non-zero angle remote from the base and away from each other; the legs and the interior side of the base defining an open channel; and (c) first and second pin slider keepers secured to the interior side of the base and overlapping a portion of the first and second through slots.
8. The clamp system of claim 6 wherein the omega clamp comprises: (a) a base with a pair of legs

extending perpendicular to the base; (i) the base defining the first and second through slots; (ii) the base having an exterior side and an opposite interior side; (b) each of the legs being bent at a non-zero angle remote from the base and away from each other: the legs and the interior side of the base defining an open channel; and (c) first and second pin slider keepers secured to the interior side of the base and overlapping a portion of the first and second through slots: (i) the first and second pin slider keepers being sized to slidably receive one of the pin flanges to put the first and second slide pin arrangements in the locked positions; and wherein when the first and second slide pin arrangements are slid to the released position, each of the pin flanges are slid to be free of the first and second pin slider keepers, and the omega clamp is free to be pivoted with the first and second pivot rods along the pivot axis away from covering the support tube.

9. A clamp system for releasably attaching a filter assembly to a support grid; the clamp system comprising: (a) one or more clamp rods; (b) a first elongated support member with openings, each opening being sized to fit over one of the clamp rods; (c) a first slide plate slidably mounted in the first support member; (i) the first slide plate having openings alignable with the openings in the first support member; (ii) the first slide plate having an end handle; (c) a second elongated support member with openings, each opening in the second support member sized to fit over one of the clamp rods; and (d) one or more handgrips, each handgrip constructed and arranged to engage one of the clamp rods and apply an axial force against the second support member.

10. The clamp system of claim 9 further comprising: (a) a second slide plate slidably mounted in the second support member; (i) the second slide plate having openings alignable with the openings in the second support member; and (ii) the second slide plate having an end handle.

11. The clamp system of claim 9 wherein the handgrips each comprises: (a) a head defining a recess sized to engage one of the clamp rods, and a cam surface on a portion of the head opposite of the recess; and (b) a grasping handle extending from the head.

12. The clamp system of claim 9 wherein (a) the first support member has a rectangular cross-sectional wall with opposite open ends, and the openings in the first support member extend through opposing sides of the wall; and (b) the second support member has a rectangular cross-sectional wall with opposite open ends, and the openings in the second support member extend through opposing sides of the wall of the second support member.

13. The clamp system of claim 9 wherein the first slide plate openings have a narrowed region and a wider region.

14. The clamp system of claim 10 wherein the second slide plate openings have a narrowed region and a wider region.

15. A filter system comprising: (a) a support grid having a frame with opposite upstream and downstream sides; (b) a first filter element oriented along the upstream side of the support grid; (c) a second filter element oriented on the support grid on the downstream side directly across and opposing the first filter element; and (d) a single clamp system releasably securing both the first filter element and the second filter element to the frame; (e) a plurality of clamp rods extending from the support grid; (f) a first elongated support member holding a first slide plate and positioned over the clamp rods; (i) the first support member being positioned against the second element gasket; and (g) a second elongated support member positioned over the clamp rods and against the first element gasket, such that the first element gasket is positioned between and against the first support member and second support member.

16. The filter system of claim 15 wherein: (a) the second filter element has a second element gasket pressed against the upstream side of the support grid; and (b) the first filter element has a first element gasket pressed against the clamp system.

17. The filter system of claim 15 further including a plurality of handgrips removably secured to the clamp rods and applying an axial force against the second support member.

18. The filter system of claim 15 further comprising a second slide plate being held by the second support member and positioned over the clamp rods.

19. The filter system of claim 15 wherein: (a) there are a plurality of first filter elements on the upstream side of the support grid; (b) the first filter elements comprise pocket filter elements or V-bank filter elements; (c) there are a plurality of second filter elements on the downstream side of the support grid; and (d) the second filter elements comprise V-bank filter elements.
